A Self-Hosted Legal Language Model: Four From-Scratch Architectures Adapted to Six Indian Legal NLP Tasks

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Abstract

We train four compact language models from scratch (103–139M parameters) on the official Indian legal corpus: a decoder-only Transformer (A), an encoder–decoder Transformer (B), a stacked LSTM (C) and a dilated residual TCN (D). We adapt them, without any external weights, to named-entity recognition, rhetorical roles, judgment prediction, statute identification, summarisation and prior-case retrieval. Across 38 submissions we compare full, layer-wise and bias-only fine-tuning; LoRA, LORMA, IA^3, prefix and prompt tuning; adapters; and knowledge distillation (KD).

- **Best results:** 0.806 BERTScore (pretraining), 0.491 strict Macro-F1 (NER), 0.518 Macro-F1 (rhetorical roles), 0.680 Macro-F1 (judgment), 0.304 Macro-F1 (statutes), 0.835 BERTScore (summaries) and 0.253 Micro-F1@7 (retrieval).
- **The backbone mattered:** the decoder A transferred best to every task except character-level NER.
- **Three methods mattered most:** a linear-probe-then-fine-tune schedule (LP-FT), which ended a series of training collapses; per-label decision calibration; and distillation, which twice produced a smaller student that beat its own teacher.

1 Introduction

A law firm that cannot send client documents to an outside API needs a model it can host itself. The capstone builds such a model in two stages:

- **Phase 1:** pretrain small language models from scratch on legal text.
- **Phases 2–7:** adapt them to downstream tasks under fixed rules: every experiment starts from the Phase-1 weights, LORMA [1] is compulsory, and at least one KD run is required per phase.

Compute was limited to free Kaggle T4 GPUs.

Contributions.

1. Four architecturally different legal LMs and one shared 50K byte-level BPE tokenizer (§3).
2. One adaptation framework reused across all phases (§4): Phase-1 backbone → PEFT method → task head, with chunked long documents, standardised features and LP-FT [3].
3. Controlled comparisons of nine adaptation techniques and the four backbones on six tasks (§5), with data-driven design choices (§2) and validation estimates that tracked the hidden test scores (Fig. 2).
4. An account of what failed and why (§7), including a four-run collapse of LORMA and its diagnosis.

Every notebook, phase README, design decision and score is in the GitHub repository above. Appendix A lists all 38 submissions.

Ph.	Data (train / dev / test)	Key finding → design
1	4.34B tok. (4 sources)	91% from high courts and tribunals → 25% equal mix
2	70 / – / 35 docs, 8.4k spans	96% of gold starts fall inside a BPE token → character-level boundaries
3	80 / 10 / 10 docs, 17k sents	context matters; 0.05–28% label skew → sentence encoder + document BiLSTM-CRF
4	31.6k / 1,988 / 1,517	**test** ⊂ **dev** → "dev-clean"; outcome cues at the end → last chunks
5	42.8k / 10.2k / 13.0k, 100 labels	42% of dev label sets unseen → binary relevance + per-statute thresholds
6	7,030 / – / 100	content spread evenly over the case → chunk-level summaries
7	827 / 118 / 237 queries	every query is in its own pool; BM25 on the full text is weak → exclude self; first and last chunks

Table 1: Task data and the decision each analysis led to.

2 Data and what it implied

Each phase began with a data-analysis notebook; its outputs are in the repository. Table 1 summarises the task data and the finding that shaped each design.

Two findings protected the scores from leakage or a trivial loss:

- In Phase 4, all 1,517 test cases also appear in dev *with labels*. We never trained or tuned on them: selection used the 1,988 remaining dev cases.
- In Phase 7, every query sits in its own candidate pool. Without exclusion, the query itself would be every query's first answer.

Other choices that followed from the analysis:

- **Casing:** train and test casing differed in Phase 4, and the tokenizer is cased, so we lower-cased there and nowhere else.
- **Placeholders:** masked placeholders (<SECTION>, <CITATION>, <ORG>) were mapped to plain words; they cost 3–5 BPE tokens each.

3 Foundation models (Phase 1)

Tokenizer. Our own byte-level BPE (vocabulary 50,000, trained on 25M characters per source); a GPT-2 tokenizer would have been trained on web text.

Training.

- Every model saw **100M tokens**, sampled equally from acts, high courts, the Supreme Court and tribunals.

	Architecture	Params	Ctx	PPL	BERTSc.
A	decoder, 12L, d=768	125.0M	2048	**20.3**	0.793
B	enc–dec, 6+6L	139.3M	1k+1k	31.8	**0.806**
C	LSTM, 3×1280	103.4M	1024	48.0	0.802
D	TCN, 10 blocks	114.2M	2048	48.2	0.805

Table 2: Phase-1 models. PPL: held-out next-token perplexity. BERTScore F1: Codabench, 128-word continuations of 512-word prompts.

- AdamW, learning rate 3×10^{-4} with cosine decay, 16k tokens per step, one T4, bf16.
- Checkpoint selection on a 2% hash-split held-out set.

Results (Table 2).

- A has by far the lowest perplexity but the lowest BERTScore.
- The four models were decoded differently: A and C with near-greedy sampling (temperature 0.1), B and D at 0.7–0.8. Near-greedy decoding favours repetition.
- BERTScore is also compressed: all four scores lie within 0.013.
- Perplexity turned out to be the better predictor of downstream value: A became the strongest backbone in Phases 3–7.

4 Adaptation framework

Every submission is *Phase-1 backbone* + *adaptation technique* + *task head*. Each notebook is self-contained and resumes from checkpoints.

Techniques.

- Full, layer-wise and bias-only (BitFit [8]) fine-tuning.
- LoRA [2].
- **LORMA** [1]: a multiplicative low-rank update $W_{\text{eff}} = (I + \frac{\alpha}{r} BA)W_0$ with $B = 0$ at the start. We used $r = 8$ and $\alpha = 16$ on q, k, v, out, fc1 and fc2 of A, and on the recurrent gates of C.
- IA^3 [4], prefix [5] and prompt tuning [6], and Houlsby / bottleneck adapters [7].
- KD [9].

Long documents.

- Cases are split into 510-token chunks. A chunk becomes [mean; final state] of its token states.
- Chunks are pooled per case with attention, mean or max. Positional chunk selection (first or last chunks) follows each task's data analysis.

Feature standardisation. A's residual stream has a few nearly constant dimensions ($|x| \approx 46$ against a mean of 2) that swamp any normalisation. We therefore standardise token or chunk features per dimension, using statistics from the frozen model. Phase 4 showed this matters (§7).

LP-FT [3].

- First a linear probe (or a retrieval projection) is trained on cached frozen features and copied into the model as "epoch 0".
- Then the adapter and head are fine-tuned gently.
- The best of epochs $0, \ldots, E$ on validation is kept, so a run can never end below its probe.

KD. Students are smaller Phase-1 models, e.g. every other layer of A (as in DistilBERT [10]). They learn from the teacher's soft labels, logits or similarity distributions (temperature 2) plus the gold loss.

5 Experiments

Table 3 lists the submissions counted in each phase. Appendix A lists every run, and Appendix B the settings.

Phase 2 (NER).

- Byte-level BPE attaches the leading space to the following token (ĠCOURT), so 97% of gold starts sit one character inside a token. We therefore predict start and end per *character*: each character gets its token's state plus a character embedding. Decoding with perfect boundary probabilities reproduces 100% of the spans.
- C+LORMA won by a wide margin.
- Calibrating span-length caps and thresholds per label added +0.10 (0.392→0.494).
- Full fine-tuning of A had not converged after 903 steps at learning rate 10^{-5}.

Phase 3 (rhetorical roles).

- **Context:** a document BiLSTM over sentence vectors roughly doubles Macro-F1 over sentence-only probes (A: 0.283→0.440; Fig. 3). A CRF adds another +0.02 to +0.04.
- **LoRA beat prefix tuning** in an otherwise identical pipeline (0.502 vs 0.453).
- **IA^3 underfitted** the weakest backbone.
- **The KD student beat its teacher** (0.518 vs 0.502) with 66% of the parameters, 1.5× faster and at half the GPU memory.

Phase 4 (judgment prediction).

- The last two chunks carry the outcome signal: first two chunks 0.593 vs last two 0.695.
- A+LORMA collapsed to a constant prediction four times. LP-FT fixed it: the probe alone gave 0.606, and LORMA fine-tuning added +0.088 (§7).
- The distilled two-layer LSTM again beat its teacher (0.592 vs 0.562).

Phase 5 (statutes).

- Per-statute thresholds were tuned with cross-fitting, i.e. tuned on one half of dev and scored on the other. Thresholds tuned and scored on the same dev data overstate Macro-F1 by about 0.015.
- The cross-fitted estimates landed within ±0.01 of test for all four runs.
- Per-statute F1 rises with training frequency (quartile 1 0.20 → quartile 4 0.45).
- The typical error is a specific statute replaced by a frequent, generic one.

Phase 6 (summarisation).

- Summarising every chunk and joining the results beats summarising the first chunk alone by +0.16 ROUGE-1.
- Changing only the decoding (minimum length 22% of the chunk plus a CTRL repetition penalty [11] of 1.2) raised BERTScore by +0.006 to +0.008 and ROUGE-1 by +0.04 to +0.08 for *every* model, without retraining (Fig. 4). Repeated trigrams fell from 20% to 11%.
- The C student distilled from A beat the directly adapted B and D.

Phase 7 (prior-case retrieval).

Phase (metric)	Sub.	Backbone + technique (head / key choice)	Trainable	Validation	**Test**
2 NER	S1	A full fine-tuning (char-level start/end head)	125.6M	0.225	0.192
(strict Macro-F1)	S2	B encoder + LoRA (QKV, then +out/FFN)	0.91M	0.198	0.177
	S3	C + **LORMA** on recurrent weights; span caps + per-label thresholds	0.79M	0.494	**0.491**
	S4	KD: D teacher (4 blocks tuned) → 6-block D student	38.4M	0.220	0.253
3 Rhetorical roles	S1	C + **LORMA** + document BiLSTM	3.39M	≈0.40	0.402
(Macro-F1)	S3	B encoder + IA^3 + document BiLSTM	2.40M	0.219	0.236
	S4	D + prompt tuning (20 tokens) + document BiLSTM	2.66M	0.329	0.339
	S7	A + LoRA (attn+FFN) + BiLSTM-CRF; stage 2 on train+dev	3.70M	0.450	0.502
	S6	KD: S7 → 6-layer A student (66% of params)	3.04M	0.458	**0.518**
4 Judgment	S1	A + **LORMA**, LP-FT, last 2 chunks	1.33M+head	0.695	**0.680**
(Macro-F1)	S2	C layer-wise FT (top LSTM layer)	13M	0.562	0.562
	S3	B encoder + BitFit	—	0.531	0.491
	S4	D + bottleneck adapters	—	0.578	0.527
	S5	KD: S2 → 2-layer C student	—	0.613	0.592
5 Statutes	S1	A + **LORMA**, LP-FT, per-statute thresholds	1.63M	0.298	**0.304**
(Macro-F1)	S2	C layer-wise FT	13.6M	0.273	0.263
	S3	D + bottleneck adapters	1.73M	0.271	0.273
	S4	KD: S1+S2+S3 ensemble → B encoder + LoRA	0.90M	0.215	0.220
6 Summaries	S1	A + **LORMA** prefix-LM, chunk summaries; tuned decoding	—	R1 0.521	**0.835**
(BERTScore F1)	S2	B + LoRA on all projections; tuned decoding	—	—	0.809
	S3	D + Houlsby adapters; tuned decoding	—	—	0.814
	S4	KD: S1 → C (word-level KL); tuned decoding	—	—	0.816
7 Retrieval	S1	A + **LORMA** bi-encoder, InfoNCE + TF-IDF hard negatives	1.72M	0.269	**0.253**
(Micro-F1@7)	S2	C layer-wise FT bi-encoder	13.8M	0.217	0.180
	S3	D + prompt tuning bi-encoder	0.55M	0.198	0.181
	S4	KD: S1 → 6-layer A (listwise KL + teacher negatives)	1.06M	0.280	0.237

Table 3: Counted submissions per phase. Validation scores never use test labels: Phase 4 uses dev-clean, Phase 5 cross-fitted thresholds, and Phase 3 the 13-label average. "—": not recorded. The best result per phase is in bold.

Ph.	Controlled change (everything else fixed)	Δ (val.)
2	per-label span caps + thresholds (S3)	+0.102
3	+ document BiLSTM vs sentence-only (A, S2)	+0.158
3	Viterbi CRF vs argmax (S7)	+0.042
3	prefix tuning vs LoRA, same pipeline (test)	−0.049
4	last 2 vs first 2 chunks (S1)	+0.102
4	LORMA fine-tuning over its linear probe (S1)	+0.088
5	first chunk only vs evidence chunks (S1)	−0.035
5	thresholds scored on their own tuning data	+0.015†
6	all chunks joined vs first chunk (ROUGE-1, S1)	+0.158
6	tuned decoding, same weights (ROUGE-1, S1)	+0.060
7	contrastive projection on frozen C features	+0.074
7	mean vs max pooling over chunks (S1)	+0.089
7	first 8 + last 8 vs first chunk only (S1)	+0.157

Table 4: Controlled ablations (validation; Phase 6 in ROUGE-1). †An optimistic bias, which is why Phase 5 reports cross-fitted estimates.

- **Model:** a bi-encoder. Each case vector is the normalised mean of chunk vectors over the first 8 and last 8 chunks; relevance is cosine similarity.
- **Training:** InfoNCE (temperature 0.05) with TF-IDF hard negatives and in-batch negatives, and LP-FT.
- **Transfer:** frozen Phase-1 vectors already retrieve (A 0.238 on dev). The contrastive projection gives most of the gain.
- **KD:** the 6-layer student encodes 1.99× faster with 66% of the parameters and keeps 94% of the teacher's test score.
- **Ceiling and reference:** with exactly 7 answers per query, an oracle reaches only 0.62 on dev. All four models stay below the TF-IDF reference (0.368 dev), but their hits are partly different from TF-IDF's: 393 relevant cases found by either vs 331 by TF-IDF alone.

6 Cross-phase analysis

What moved the scores (Table 4). The largest controlled gains came from task structure rather than from the adapter itself: which part of the document is read, document-level context, and calibrating decisions or decoding. Within a fixed backbone, the adapter changed scores by 0.02–0.09.

Which backbone transfers (Fig. 1).

- The decoder A is best in Phases 3–7, and frozen-feature probes predict this ranking before any training.
- C wins only NER, where boundaries are local and recurrence suits character-level decisions.
- The encoder of B was consistently weakest when frozen. Its decoder was trained to read it, not to be a feature extractor.

PEFT vs full adaptation. On small datasets (70 NER documents, 80 rhetorical-role documents, 817 retrieval queries), PEFT beat or matched heavier tuning:

- full fine-tuning of 125M parameters lost to LORMA on 0.8M (NER);
- layer-wise tuning of 13M parameters tied prompt tuning on 0.55M (retrieval).

Capacity still has a floor: IA^3 (27.6K parameters) and BitFit underfitted. Techniques that *add new directions* to the weights (LoRA, LORMA) beat those that only rescale (IA^3, BitFit) or only add inputs (prefix, prompt).

Does KD preserve quality? Yes, in all six phases. Students beat their teachers twice (rhetorical roles, judgment) and kept 94% in retrieval. In a fourth phase the distilled LSTM beat two directly adapted models (summaries). Students lost when they came from a weak backbone: the B encoder kept 69% in Phase 5. KD worked as a regulariser on small, noisy data as much as as compression.

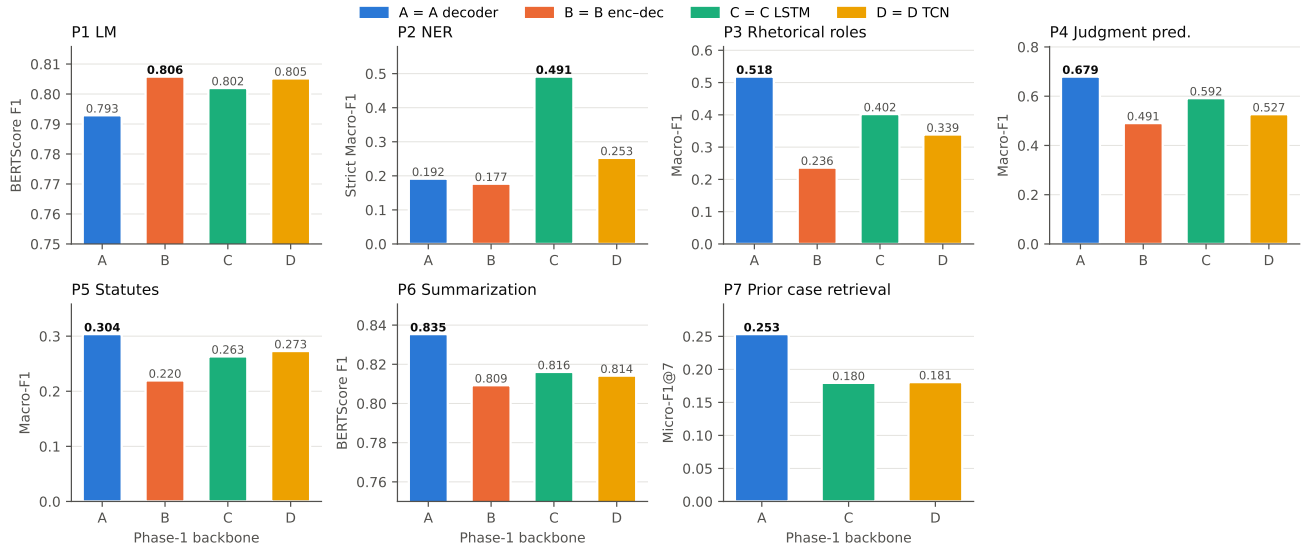


Figure 1: Best test score per backbone and phase. A wins every downstream phase except NER (C).

Were the validation estimates reliable? (Fig. 2). Leak-free validation ranked submissions as the test did in every phase. The largest gaps came from the smallest dev sets (10 documents in Phase 3, 118 queries in Phase 7).

Cost vs score. The best runs were not the most expensive per point:

- LORMA on A trained only 1.3–1.7M parameters, at 2–9 GB of GPU memory and 45–160 minutes per run.
- Decoding changes in Phase 6 cost no training at all.

7 What failed and why

LORMA collapse (Phase 4). A+LORMA predicted one class for every case in four consecutive runs. We changed one suspected cause at a time:

1. Attention pooling: no effect.
2. Standardising token states: the diagnostic notebook found nearly constant outlier dimensions. Still collapsed.
3. Turning off dropout in the frozen backbone: dropping one outlier dimension shifts LayerNorm for the whole token. Still collapsed.

Final cause: with a randomly initialised head, the first multiplicative updates wreck the features before the head finds the weak signal. LP-FT solved it, and the same schedule then worked in Phases 5 and 7.

Other failures.

- **Phase 1 decoding:** near-greedy sampling for A and C (repetition).
- **Phase 2:** full fine-tuning of A at a learning rate that was too low; a "LORMA" that was in fact additive, replaced by the true multiplicative form.
- **Phase 3:** the TCN's dilated convolutions hit a slow GPU path (220 minutes per epoch); rewriting them as exact per-tap matrix multiplies made them about 120× faster.
- **Phase 6:** the original decoding was too short and repetitive.
- **Phase 7:** the neural rankers stayed below TF-IDF. A next-token objective over 100M tokens does not encode *which legal issue* a case is about, and 817 queries with roughly 15 in-batch negatives cannot reshape that much.

8 Limitations and future work

Limitations.

- Each Phase-1 model saw only 2.3% of the corpus (100M of 4.3B tokens); all were still improving except A.
- Most runs were limited to 2–4 epochs by Kaggle session limits.
- Several validation sets were tiny (10 documents, 118 queries).
- Peak-memory figures are missing for Phases 1 and 4, and the Phase-7 KD memory comparison did not separate the two models.

Future work.

- Pretrain longer with a retrieval-aware objective (contrastive pairs from citations).
- Use larger in-batch negative pools via gradient caching.
- Combine neural and lexical scores in Phase 7, which the complementary hits suggest.
- Use a single A-family student, distilled once, for all tasks.

9 Conclusion

A self-hosted 125M-parameter decoder trained on 100M legal tokens, adapted with about 1% extra parameters through LORMA and LP-FT, reached the best result in five of six downstream tasks. Distilled students of 66% size kept, and twice exceeded, their teachers' quality. The largest gains came less from the choice of adapter than from:

- reading the data correctly (leaks, token boundaries, which part of a document matters);
- stable optimisation (standardisation, LP-FT);
- calibrating decisions (thresholds, decoding).

LLM usage declaration

[Replace with the questionnaire from the official template.]

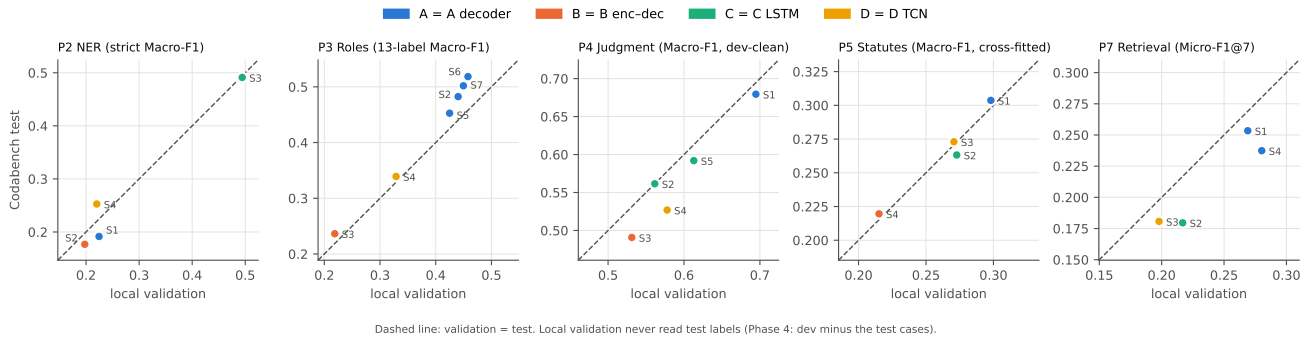


Figure 2: Local validation vs Codabench test, for every submission with a comparable validation score.

- **Tool:** Claude (Anthropic) was used as a coding and writing assistant.
- **What it did:** drafting the Kaggle notebooks from our specifications, debugging (e.g. the Phase-4 collapse diagnosis), writing the READMEs, and drafting this report.
- **What it did not do:** it used no project data and produced no model outputs. No LLM or API was used inside any submitted model, and no pretrained weights were used.
- **What we did:** all notebooks were run, checked and submitted by the author, and all numbers come from our own runs.

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A All submissions

Every Codabench run, from `results/submissions.csv`. The *Kept* column marks the runs on the final leaderboard. Replaced runs: Phase 3 S2 and S5 (now ablations: LoRA without a CRF; prefix tuning vs LoRA), the two earlier Phase-4 S1 runs (superseded by the final LP-FT run), and the original Phase-6 decoding (superseded by the tuned decoding of the same weights).

Ph.	Notebook	Base model / technique	Trainable	Local validation	Test	Kept
1	A_decoder_only	A Decoder-only: from-scratch pretraining	125.03M	val PPL 20.30	0.7929	yes
1	B_Encoder_Decoder	B Encoder-Decoder: from-scratch pretraining	139.26M	val PPL 31.82	0.8058	yes
1	C_StackedLSTM	C Stacked LSTM: from-scratch pretraining	103.40M	val PPL 47.97	0.8020	yes
1	D_TCN	D TCN: from-scratch pretraining	114.16M	val PPL 48.17	0.8052	yes
2	S1_Decoder_FullFineTuning	A Decoder-only: full fine-tuning	125.64M	0.2245	0.1916	yes
2	S2_Encoder_Decoder_LORA	B Encoder-Decoder: LoRA / expanded LoRA	0.91M	0.1976	0.1768	yes
2	S3_StackedLSTM_LORAM	C Stacked LSTM: LORMA	0.79M	0.4943	0.4912	yes
2	S4_TCN_KnowDist	D TCN: knowledge distillation	38.45M	0.2202	0.2527	yes
3	P3_S1_C_LSTM_LORMA	C Stacked LSTM: LORMA r=8 + doc BiLSTM	3.39M	0.4762 (dev; CL 0.3902 IT 0.4695)	0.4024	yes
3	P3_S2_A_Decoder_LoRA	A Decoder-only: LoRA r=8 q/k/v/out + doc BiLSTM	2.96M	0.4402 (dev 13-label; present 0.5202)	0.4825	no
3	P3_S3_B_Encoder_IA3	B Encoder-Decoder (encoder): IA3 keys/values/FFN + doc BiLSTM	2.40M	0.2188 (dev 13-label; present 0.2586)	0.2364	yes
3	P3_S4_D_TCN_PromptTuning	D Dilated Residual TCN: Prompt tuning 20 tokens + doc BiLSTM	2.66M	0.3289 (dev 13-label; present 0.3886)	0.3391	yes
3	P3_S7_A_LoRA_CRF_TrainDev	A Decoder-only: LoRA attn+FFN + BiLSTM-CRF; Stage 2 train+dev	3.70M	0.4497 (Stage-1 dev 13-label Viterbi; argmax 0.4082)	0.5019	yes
3	P3_S5_A_Prefix_CRF_TrainDev	A Decoder-only: Prefix tuning 20x12 (MLP) + BiLSTM-CRF; Stage 2 train+dev	12.24M	0.4249 (Stage-1 dev 13-label Viterbi; argmax 0.4063)	0.4527	no
3	P3_S6_KD_S7Teacher_6LayerStudent	A 6 of 12 layers (student): KD from S7 (T=2) + LoRA + BiLSTM-CRF; Stage 2 train+dev	3.04M	0.4579 (student Stage-1 dev; teacher 0.4497)	0.5184	yes
4	P4_S1_A_LORMA	A Decoder-only: LORMA r=8 attn+FFN; last 2x510 chunks; standardised states	–	0.3452 (dev-clean; collapsed)	0.3475	no
4	P4_S2_C_LSTM_LayerwiseFT	C Stacked LSTM: Layer-wise FT: top LSTM layer + final norm; last 4 chunks at test	–	0.5615 (dev-clean last 4)	0.5615	yes
4	P4_S3_B_Encoder_BitFit	B Encoder (6L): BitFit bias-only	–	0.5312 (dev-clean)	0.4908	yes
4	P4_S4_D_TCN_Adapters	D Dilated Residual TCN: Bottleneck adapters 1024-64-1024; last 4 chunks at test	–	0.5778 (dev-clean last 4)	0.5269	yes
4	P4_S5_KD_S2Teacher_C2LayerStudent	C 2 of 3 LSTM layers (student): KD from S2 (T=2) + layer-wise FT; last 8 chunks at test	–	0.6129 (dev-clean last 8; teacher 0.5615)	0.5920	yes
4	P4_S1_A_LORMA	A Decoder-only: LORMA r=8 attn+FFN with LP-FT (probe init then FT); last 2 chunks	–	0.6710 (dev-clean last 2; epoch 2)	0.6732	no
4	P4_S1_A_LORMA	A Decoder-only: LORMA r=8 attn+FFN with LP-FT; 4 epochs; last 2 chunks	–	0.6947 (dev-clean last 2; epoch 3 of 4)	0.6795	yes
5	P5_S1_A_LORMA	A Decoder-only: LORMA r=8 LP-FT; evidence chunks; per-statute thresholds	1.63M	0.2979 (dev cross-fitted)	0.3037	yes
5	P5_S2_C_LSTM_LayerwiseFT	C Stacked LSTM: Layer-wise FT (top layer) LP-FT	13.63M	0.2728 (dev cross-fitted)	0.2632	yes
5	P5_S3_D_TCN_Adapters	D Dilated Residual TCN: Bottleneck adapters LP-FT	1.73M	0.2707 (dev cross-fitted)	0.2730	yes
5	P5_S4_KD_Ensemble_to_BEncoder_LoRA	B encoder (student): KD from S1+S2+S3 ensemble + LoRA	0.90M	0.2152 (dev cross-fitted)	0.2196	yes
6	P6_S1_A_LORMA	A Decoder-only: LORMA r=8; chunk-level prefix-LM summaries joined	–	–	0.8287	no
6	P6_S2_B_EncDec_LoRA	B Encoder-Decoder: LoRA r=8 all enc/dec projections; seq2seq chunk summaries joined	–	–	0.8033	no
6	P6_S3_D_TCN_HoulsbyAdapters	D Dilated Residual TCN: Houlsby adapters; prefix-LM chunk summaries joined	–	–	0.8060	no
6	P6_S4_KD_A_to_C_LSTM	C Stacked LSTM (student): Word-level KD from S1 (A + LORMA) T=2	–	–	0.8086	no
6	P6_S1_tune_generation	A Decoder-only: S1 weights; decoding min length 22% + repetition penalty 1.2	–	0.5212/0.2495 (val ROUGE-1/L)	0.8354	yes
6	P6_S2_tune_generation	B Encoder-Decoder: S2 weights; tuned decoding	–	–	0.8093	yes
6	P6_S3_tune_generation	D Dilated Residual TCN: S3 weights; tuned decoding	–	–	0.8142	yes
6	P6_S4_tune_generation	C Stacked LSTM (KD student): S4 weights; tuned decoding	–	–	0.8161	yes
7	P7_S1_A_LORMA	A Decoder-only: LORMA bi-encoder, LP-FT, InfoNCE + TF-IDF hard negatives	1.72M	0.2690 (dev Micro-F1@7)	0.2534	yes
7	P7_S2_C_LSTM_LayerwiseFT	C Stacked LSTM: Layer-wise FT (top LSTM layer + final norm) bi-encoder, LP-FT	13.78M	0.2168 (dev)	0.1796	yes
7	P7_S3_D_TCN_PromptTuning	D Dilated Residual TCN: Prompt tuning (20 soft tokens) bi-encoder, LP-FT	0.55M	0.1979 (dev)	0.1807	yes
7	P7_S4_KD_S1Teacher_A6LayerStudent	A 6 of 12 layers (student): KD from S1 (listwise KL + teacher-mined negatives) + LORMA	1.06M	0.2802 (dev; teacher 0.2690)	0.2374	yes

B Settings shared by most runs

- **Optimiser:** AdamW, weight decay 0.01, linear warm-up over 6% of steps then linear decay, gradient clip 1.0, fp16 autocast.
- **Adapters:** LORMA and LoRA with $r=8$, $\alpha=16$. Learning rate 10^{-4} for LORMA in LP-FT, 3×10^{-4} for LoRA, 3×10^{-3} for IA^3, 10^{-2} (Phase 3) or 3×10^{-3} (Phase 7) for prompts.
- **Chunks:** 510 BPE tokens plus [BOS]/[EOS].
- **Phase 3:** windows of 128 sentences, stride 64; class weights $(N/Kn_c)^{0.5}$, capped at 10.
- **Phase 5:** BCE with positive weight $\sqrt{\text{neg}/\text{pos}}$, capped at 10.
- **Phase 7:** InfoNCE at temperature 0.05; 4 query–positive pairs per step, each with 2 hard negatives; projection to 256 dimensions; KD at temperature 2 with weight 1.0.

• **KD students:** every other layer of A (Phases 3 and 7), 2 of 3 LSTM layers (Phase 4), 6 of 10 TCN blocks (Phase 2). Exact values for every run are in the first settings cell of each notebook and in `docs/design_decisions.md`.

C Additional analysis

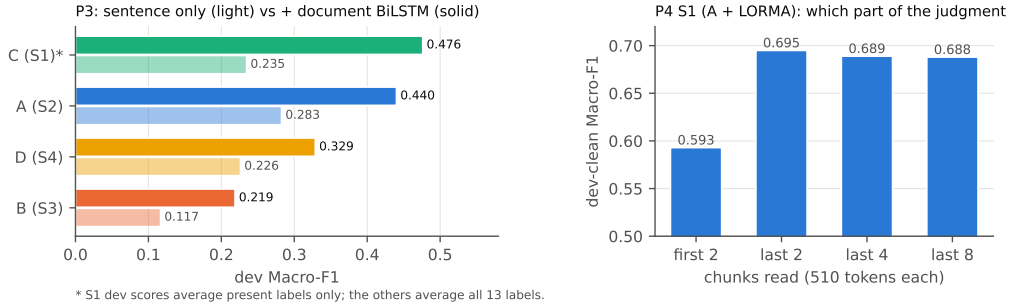


Figure 3: Left: Phase-3 document context (sentence-only probe vs document BiLSTM). Right: Phase-4 section analysis for A+LORMA. C's 0.476 averages only the labels that appear in dev; the others average all 13 labels.

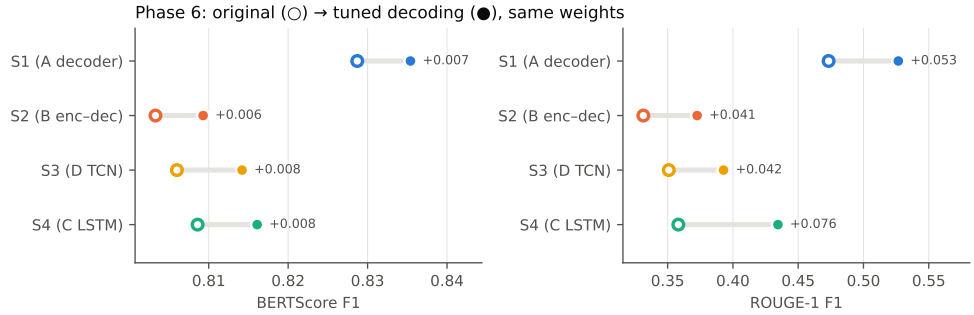


Figure 4: Phase 6: the same weights with the original vs the tuned decoding, for all four models.

Phase-7 KD report (100 dev documents, first 4 + last 4 chunks, one T4).
- **Teacher:** 126.7M parameters, 7.04 s to encode, dev 0.269, test 0.2534.
- **Student:** 83.6M parameters, 3.54 s, dev 0.280, test 0.2374.
- **Comparison:** top-7 agreement with the teacher was 49%. The teacher rebuilt from its checkpoint reproduces the saved vectors (cosine 1.000).

Phase-7 error pattern.
- Queries with one relevant case reach precision@7 of only 0.05–0.08.
- Queries with 16 or more relevant cases reach precision 0.38–0.56, but recall only 0.09–0.13.
- S1 found no relevant case for 18 of 118 dev queries.

Per-phase details. Each `code/phaseN/README.md` has the full data analysis, per-class results, error examples and Kaggle links for its phase. The experiment log in `docs/` records every run in order.